Verifiable Reasoning Framework for Legal Question Answering via RAG and Symbolic Rules

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***Abstract*—Legal Question Answering (QA) requires the ability to correctly apply legal provisions and derive conclusions supported by explicit legal grounds. Retrieval-Augmented Generation (RAG) improves access to relevant legal documents but primarily relies on generating answers from retrieved passages. This paper proposes a framework for Vietnamese Legal Question Answering that integrates large language models (LLMs) with a structured Legal Knowledge Base and a verifiable reasoning process. Legal provisions are transformed into structured legal rules that support rule retrieval and symbolic reasoning to verify answer generation. Through experimental results on four legal domains, the proposed framework improves the consistency and reliability of legal question answering by explicitly separating knowledge representation, reasoning, and verification.**

***Index Terms*—Legal Question Answering, Knowledge Base, Large Language Model, Retrieval-Augmented Generation, Neuro-Symbolic Reasoning**

I. Introduction

Legal Question Answering (QA) plays an important role in legal consultation and decision support systems [1], [2]. Unlike general question answering, a legal answer must not only be factually correct but also be supported by appropriate legal provisions, satisfy all applicable legal conditions, and provide an explainable reasoning process [3], [4]. In practice, legal regulations are often distributed across multiple legal documents, while user queries may contain incomplete information or involve multiple legal domains simultaneously.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have significantly improved language understanding and legal document retrieval [5], [6]. However, retrieving relevant legal documents alone is insufficient to guarantee legally valid conclusions if the applicable legal conditions are not systematically identified and verified before reasoning [7]. Many legal cases require integrating regulations scattered across multiple legal documents, whereas existing RAG-based systems primarily focus on document retrieval [8]. Meanwhile, neuro-symbolic reasoning provides a structured approach to knowledge representation and multi-step reasoning over heterogeneous knowledge sources [9], [10].

This paper presents an integrated neuro-symbolic framework for Vietnamese Legal Question Answering rather than introducing a new general-purpose reasoning algorithm. The framework combines an LLM-based semantic parser, a structured Legal Knowledge Base, explicit forward reasoning, proof construction, and source-grounded verification. Its primary contribution lies in integrating neural language understanding with provenance-preserving symbolic reasoning to produce legally grounded and traceable answers. An empirical evaluation on 443 Vietnamese legal questions spanning four legal domains shows that the proposed framework achieves higher IRAC scores than Qwen2.5-7B-Instruct and GPT-4o, particularly on cross-domain questions requiring reasoning over multiple legal provisions.

II. Related Work

Neuro-symbolic reasoning combines the representation learning capability of neural networks with the interpretability and logical consistency of symbolic reasoning [11]. Representative approaches, including Markov Logic Networks (MLNs) [12], Logic Tensor Networks (LTNs) [13], [14], and DeepProbLog [15], have demonstrated the effectiveness of integrating neural learning with symbolic inference across a wide range of reasoning tasks. More recently, retrieval-enhanced frameworks such as Retrieval-Augmented Generation (RAG) [5], [16], REALM [6], and Self-RAG [17] have improved the factual reliability of large language models by incorporating external knowledge during inference. However, these approaches generally retrieve textual evidence rather than performing explicit symbolic reasoning over structured legal knowledge.

Legal QA has also attracted increasing attention in recent years. Benchmark datasets such as LexGLUE [18] and LegalBench [19] have facilitated the evaluation of legal language understanding and legal reasoning tasks. In parallel, domain-specific pre-trained language models, such as LEGAL-BERT [20], have demonstrated the effectiveness of adapting language models to legal corpora. Nevertheless, most existing Legal QA systems primarily rely on retrieving legal

text passages and generating answers directly from retrieved documents, leaving the reasoning process and verification of legal conclusions largely implicit [8], [20].

Dang et al. [8] combine Legal-Onto, knowledge graphs, RAG, and GPT-4o, using semantic triples to retrieve legal documents for answer generation. In contrast, our framework represents provisions as provenance-preserving Horn rules, retrieves candidate rules, performs forward reasoning, constructs an explicit proof, and verifies the result against original provisions. Direct numerical comparison is inappropriate because their study evaluates 197 questions using retrieval and answer accuracy, whereas ours evaluates 443 questions across four domains using IRAC.

III. Legal Knowledge Organization

This section describes how Vietnamese legal documents are transformed into structured rules and organized in the Legal Knowledge Base for symbolic reasoning, efficient retrieval, and traceability.

A. Legal Knowledge Base

To support symbolic legal reasoning, official Vietnamese legal documents are transformed into a structured Legal Knowledge Base composed of legal rules rather than unstructured text passages, enabling explicit symbolic reasoning and legal traceability [21]. As illustrated in Figure 1, the Legal Knowledge Base covers four legal domains while preserving the hierarchical organization of Vietnamese legal documents into chapters, articles, clauses, and points in accordance with the Law on Promulgation of Legal Documents [22].

Vietnamese legal provisions often combine prerequisite conditions, exceptions, temporal constraints, and cross-references, making isolated passage retrieval insufficient. Accordingly, the Legal Knowledge Base preserves hierarchy and provenance, represents conditions and consequences as Horn-rule body and head predicates, and stores exceptions and temporal constraints as rule attributes. Semantic parsing extracts the legal subject, action, facts, and goal; candidate retrieval identifies relevant rules across provisions; and forward reasoning constructs a proof linking applied rules to their sources.

B. Legal Knowledge Base Construction

The Legal Knowledge Base is constructed through a four-stage pipeline consisting of document preprocessing, legal structure segmentation, legal rule extraction, and knowledge base construction, as illustrated in Figure 1. The objective of this pipeline is to transform unstructured Vietnamese legal documents into structured legal rules suitable for symbolic reasoning and legal verification.

First, legal documents are converted into plain text and preprocessed to normalize formatting while preserving their hierarchical legal structure. The documents are subsequently segmented into chapters, articles, clauses, and points, producing legal text units for rule extraction.

From these normalized legal units, a rule-based extraction process identifies normative legal provisions and determines

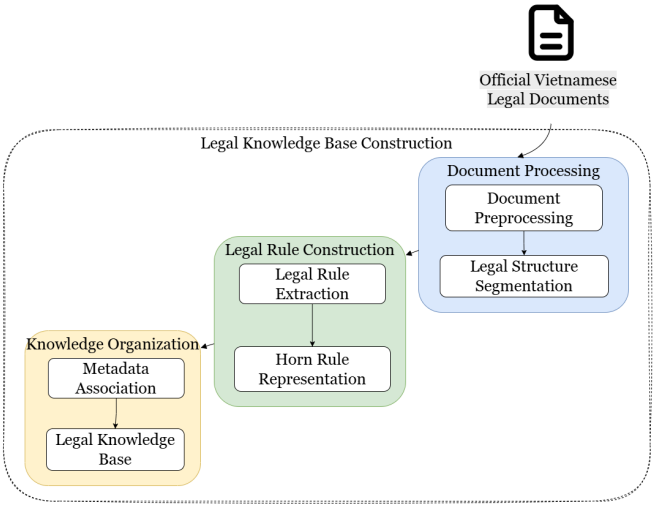


Fig. 1. Pipeline for constructing the Legal Knowledge Base.

the corresponding legal consequence together with its prerequisite legal conditions. The legal consequence is represented as the head predicate of a Horn rule, whereas the prerequisite legal conditions are represented as body predicates. Additional legal attributes, such as the applicable subject, legal action, exceptions, and temporal constraints, are retained as structured rule attributes when applicable.

Consistent with the Horn clause representation adopted in neuro-symbolic reasoning frameworks such as DeepProbLog [15], each extracted legal rule is represented as

$$r_i : \; h_i \leftarrow b_{i1},\; b_{i2},\; \ldots,\; b_{im} \tag{1}$$

where r_i denotes the i-th legal rule, h_i represents the rule head corresponding to the legal conclusion or legal effect, and b_{ij} denotes the body predicates representing the legal conditions required for applying the rule.

For example, the legal provision requiring an enterprise to register a change of its legal representative is represented as

$$r : \; \text{obligation}(X, \text{register}) \leftarrow \text{enterprise}(X), \; \text{changeRep}(X). \tag{2}$$

In Figure 1, each extracted rule is associated with structured metadata, including the source document, chapter, article, clause, point, legal domain, and rule identifier. This metadata preserves Article-Clause-Point provenance and allows every rule used during reasoning to be traced to its original legal provision. After rule extraction and metadata association, the resulting rules are incorporated into the Legal Knowledge Base to support candidate rule retrieval, forward reasoning, proof construction, and source-grounded verification.

IV. Neuro-Symbolic Reasoning

This section presents the proposed neuro-symbolic reasoning framework for Vietnamese Legal Question Answering. Built upon the structured Legal Knowledge Base introduced in

Section III, the framework combines neural language understanding with symbolic reasoning to generate legally grounded and verifiable answers.

As illustrated in Figure 2, the proposed framework consists of four sequential modules: Semantic Parsing, Candidate Rule Retrieval, Forward Reasoning, and Verification with Answer Generation. The neural component transforms a natural-language legal question into a structured legal query, while the symbolic component retrieves candidate legal rules, performs explicit forward reasoning to construct a reasoning proof, and verifies the inferred legal conclusions before generating the final legal answer. The following subsections describe each module in detail.

A. Semantic Parsing

The objective of this module is to transform a natural-language legal question into a structured legal query suitable for subsequent symbolic reasoning.

Recent advances in legal natural language understanding and retrieval-augmented reasoning employ large language models to bridge the gap between natural-language legal expressions and structured knowledge representations [5], [20]. Following this paradigm, semantic parsing in the proposed framework is performed using GPT-4o, accessed in June 2026, under a prompt-based setting without task-specific fine-tuning. Deterministic decoding is applied with temperature set to 0.0 and top-p set to 1.0.

The parser output is represented and validated against a predefined two-layer Pydantic schema. Layer 1 captures surface linguistic and legal-semantic information, including the utterance type, legal subject, conditions, action, modality, temporal information, exceptions, question focus, and assertion status. Layer 2 normalizes these elements into a subject representation, logical condition atoms, case facts, and a predicate-argument legal goal, which together form the Structured Legal Query used for candidate rule retrieval and forward reasoning.

The semantic parsing procedure consists of the following three steps.

Step 1. Identify the legal subject and legal action.

The neural language model uses the semantic context of the question to isolate the primary legal subject and action from descriptive information that is not directly relevant to legal reasoning. These elements establish the legal context of the query and serve as semantic anchors for extracting relevant case facts and retrieving candidate legal rules.

Step 2. Extract case facts.

Using the legal context established in the previous step, the neural language model extracts legally relevant facts explicitly stated in the user's question. These facts may include personal attributes, legal status, temporal information, and quantitative information that influence the applicability of legal rules. Only information explicitly supported by the question is retained, while assumptions and unsupported information are excluded. The extracted facts constitute the initial working fact set and are used together with the identified subject, action, and legal goal during candidate rule retrieval and forward reasoning.

Step 3. Determine the target legal goal.

Based on the identified legal subject, action, and case facts, the neural model determines the target legal consequence as the objective of symbolic inference. These elements are compiled into a unified structured legal query, which is then passed to the candidate rule retrieval module to retrieve relevant rules and initialize the forward reasoning process.

Errors produced during semantic parsing may propagate to subsequent modules. An incorrectly identified legal subject or action may cause candidate rule retrieval to return irrelevant rules, while omitted or incorrectly normalized case facts may prevent applicable rule premises from being satisfied. An incorrectly determined legal goal may also direct forward reasoning toward an unintended conclusion or stopping condition. Therefore, a formally valid proof over the retrieved rules does not necessarily guarantee semantic correctness when the initial structured query does not accurately represent the user's question.

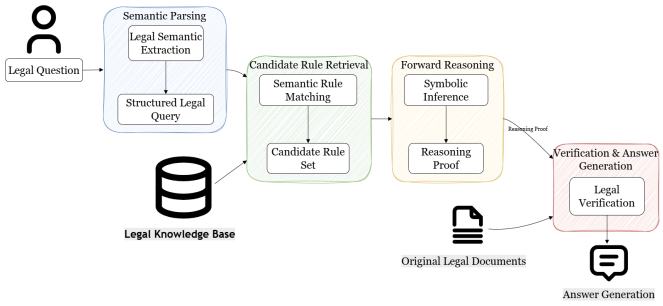


Fig. 2. Overall architecture of the proposed neuro-symbolic Legal Question Answering framework.

B. Candidate Rule Retrieval

The objective of this module is to retrieve legal rules that are semantically relevant to the user's legal question while reducing the search space for subsequent symbolic reasoning. Unlike conventional Legal Question Answering systems that retrieve unstructured legal text passages, the proposed framework retrieves structured legal rules from the Legal Knowledge Base, enabling symbolic inference to operate directly on explicit legal semantics.

The candidate rule retrieval procedure consists of the following two steps.

Step 1. Match legal rules.

The structured legal query generated by the semantic parsing module is matched against the legal rules stored in the Legal Knowledge Base. The matching process primarily compares the target legal goal together with the identified legal subject, legal action, and extracted case facts against the predicates of legal rules to retrieve semantically relevant candidates. The associated legal metadata, including the legal domain and document identifiers, is further used to filter and prioritize the retrieved rules while preserving legal traceability. The resulting candidate rule set may therefore contain legal rules originating

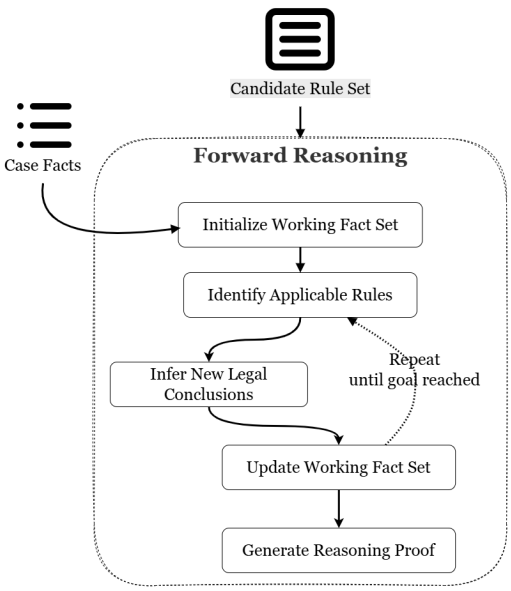


Fig. 3. Forward reasoning process for iterative symbolic inference and reasoning proof construction.

from multiple legal documents or legal domains when cross-domain reasoning is required.

Step 2. Construct the candidate rule set.

The retrieved legal rules are organized into a candidate rule set for symbolic reasoning. Each candidate rule retains both its logical representation and its associated metadata [23], [24], including the source document, chapter, article, clause, legal domain, and rule identifier. By restricting symbolic inference to this candidate rule set, the framework substantially reduces the reasoning space while preserving all semantically relevant legal rules. The resulting candidate rule set is subsequently passed to the forward reasoning module, where the target legal goal serves as the stopping criterion of the symbolic inference process.

C. Forward Reasoning

As shown in Figure 3, the module applies forward reasoning over candidate Horn rules [25]. Starting from the case facts, it iteratively applies satisfied rules and adds inferred conclusions until reaching the target goal or a fixed point, while recording the applied rules and intermediate conclusions as an explicit multi-hop proof.

The forward reasoning procedure consists of the following four steps.

Step 1. Initialize the working fact set.

The reasoning process begins by constructing the initial working fact set from the case facts extracted during semantic parsing. These case facts represent the factual circumstances of the user's legal question and constitute the initial knowledge available for symbolic inference. Simultaneously, the candidate rule set retrieved from the Legal Knowledge Base is loaded as the symbolic reasoning space, while the target legal goal specifies the legal consequence to be derived and serves as the stopping criterion of the reasoning process. The initialized working fact set provides the basis for evaluating the applicability of candidate rules during each reasoning iteration.

Step 2. Identify applicable legal rules.

During each reasoning iteration, every legal rule contained in the candidate rule set is evaluated against the current working fact set. For each candidate rule, the framework determines whether all predicates appearing in the rule body are supported by the available case facts or by legal conclusions inferred during previous reasoning iterations. Candidate rules satisfying all prerequisite conditions are regarded as applicable and selected for inference in the current iteration, whereas the remaining rules are retained because newly inferred legal conclusions may satisfy their prerequisite conditions in subsequent iterations.

Step 3. Infer new legal conclusions.

Each applicable legal rule is applied by inferring its head predicate from the satisfied body predicates. Every newly inferred legal conclusion is immediately incorporated into the working fact set, allowing additional candidate legal rules to become applicable during subsequent reasoning iterations. The inference process is repeatedly performed until the target legal goal is derived or no further legal conclusions can be inferred. By continuously expanding the working fact set through newly inferred legal conclusions, the framework naturally supports multi-hop reasoning across multiple legal rules and, when necessary, across different legal domains.

Step 4. Construct the reasoning proof.

After the inference process terminates, the framework organizes the complete reasoning trace into an explicit reasoning proof. For each inferred legal conclusion, the proof records the supporting case facts, the applied legal rule, intermediate legal conclusions derived during previous reasoning iterations, and the resulting legal consequence. By explicitly preserving the dependency between applicable legal rules, and inferred legal conclusions, the generated reasoning proof provides a transparent and traceable explanation of the entire symbolic reasoning process. The resulting reasoning proof is subsequently passed to the verification and answer generation module.

D. Verification and Answer Generation

The objective of this module is to validate the reasoning proof generated during forward reasoning before producing the final legal answer. Inspired by retrieval-augmented generation frameworks [5], the proposed framework retrieves the original legal documents as authoritative evidence for verification before answer generation. This verification layer provides an additional consistency check between the inferred conclusions and the authoritative legal sources.

The verification and answer generation procedure consists of the following four steps.

Step 1. Identify supporting legal provisions.

The reasoning proof generated during forward reasoning is first analyzed to identify all legal provisions referenced

throughout the inference process. The corresponding document identifiers, legal rule identifiers, and legal metadata are extracted to formulate retrieval queries for the original legal documents. These legal references establish the connection between the symbolic reasoning results and the authoritative legal sources used for verification.

Step 2. Retrieve the original legal documents.

Using the identified legal references, the framework retrieves the corresponding legal text, including the relevant chapters, articles, clauses, and points, from the original legal document repository. The retrieved legal text serves as the authoritative legal evidence for verification and provides an additional validation layer by comparing the symbolic reasoning results with the original legal documents.

Step 3. Verify the reasoning result.

The retrieved legal text and reasoning proof are provided to an LLM-based verifier, which checks whether the applied rules are supported by the cited provisions, whether the inferred conclusions are consistent with the legal text, and whether unsupported interpretations or contradictions are present. The resulting consistency assessment, reasoning proof, and supporting legal grounds are then used for natural-language answer generation.

Step 4. Generate the final legal answer.

The verified reasoning result, including the validated legal consequence and supporting legal provisions, is provided to a large language model to generate a natural-language legal response. During answer generation, the language model is instructed to preserve the verified reasoning result without introducing unsupported legal conclusions or modifying the validated legal grounds. The generated response integrates the validated legal conclusion with the corresponding supporting legal provisions into a coherent and readable explanation.

V. Experiments

This section presents the experimental setup and quantitatively evaluates the proposed neuro-symbolic framework for Vietnamese Legal Question Answering. Comparative experiments with representative large language model baselines are conducted to assess the effectiveness of the proposed approach.

A. Experimental Setup

The Legal Knowledge Base was constructed from 28 official Vietnamese legal documents covering social insurance, enterprise, taxation, and cross-domain cases using the pipeline in Section III. After preprocessing, it contains 207 chapters, 3,160 articles, 5,417 clauses, and 7,167 points. The extraction pipeline produced 8,020 Horn rules, each linked to its document, chapter, article, clause, and point, thereby preserving provenance for retrieval, reasoning, and verification.

The evaluation dataset comprises 443 real-user questions collected from publicly accessible Vietnamese legal consultation websites, together with the corresponding lawyer-provided answers and legal grounds.

The proposed framework was compared with Qwen2.5-7B-Instruct and GPT-4o using the same prompts and deterministic decoding ($temperature = 0.0$, $top\text{-}p = 1.0$).

Following prior IRAC-based evaluations [19], [26], answers were assessed across five dimensions: Issue (I), Rule (R), Analysis (A), Conclusion (C), and Legal Grounding (L), together with an overall IRAC score. A single annotator manually evaluated each generated answer against the corresponding lawyer-provided answer and legal grounds, which served only as references rather than a second independent rating.

B. Experimental Results

Table I summarizes the IRAC performance of the evaluated systems across the four legal domains.

TABLE I
IRAC PERFORMANCE ACROSS LEGAL DOMAINS

Domain	Cases	Qwen	GPT-4o	Our method
Social Insurance	196	50.66	60.68	**80.92**
Enterprise	119	50.87	59.01	**71.22**
Taxation	69	49.64	61.90	**68.88**
Cross-domain	59	38.01	47.43	**77.82**
Overall	443	48.87	58.66	**76.03**

Table I and Figure 4 demonstrate that the proposed framework consistently outperforms both baselines across all domains, achieving the highest overall IRAC score of 76.03. The largest gain occurs on Cross-domain questions, where it substantially exceeds GPT-4o and Qwen2.5-7B-Instruct, indicating that explicit symbolic reasoning benefits multi-provision integration. Among single-domain categories, Social Insurance yields the highest performance, while Enterprise and Taxation remain more challenging. This suggests that diverse legal conditions and multiple regulations present greater reasoning difficulty, even with structured knowledge.

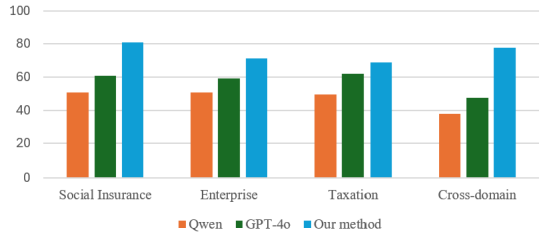


Fig. 4. Comparison between the systems on each domain.

GPT-4o generated fluent and plausible answers, while Qwen2.5 handled straightforward questions effectively. Both baselines frequently omitted essential legal conditions, cited incomplete grounds, or drew premature conclusions. These limitations became particularly evident for Cross-domain questions requiring legal knowledge from multiple regulations.

The proposed framework addresses these limitations by explicitly separating legal rule retrieval, symbolic reasoning, verification, and answer generation. Instead of relying on implicit language model reasoning, legal conclusions are derived only after satisfying all required conditions during forward reasoning. Consequently, every generated answer is

supported by explicit legal grounds and a traceable process, enhancing consistency and reliability. These findings suggest the framework's primary contribution lies not in stronger language generation, but in augmenting language models with explicit legal reasoning capabilities and legal traceability.

VI. Conclusion and Future Work

This paper presented an integrated neuro-symbolic framework for Vietnamese Legal QA that combines semantic parsing, provenance-preserving Horn rules, symbolic reasoning, proof construction, and source-grounded verification. On 443 questions across four domains, it outperformed Qwen2.5-7B-Instruct and GPT-4o in IRAC scores, particularly for cross-domain reasoning. Limitations include single-annotator scoring without inter-rater reliability, no separate annotation of question completeness or ambiguity, and the potential propagation of semantic parsing errors to downstream reasoning stages. Future work will address these limitations, expand the knowledge base, improve multi-step reasoning, and automate legal updates.

Acknowledgments

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